

# Python Collections – Detailed Cheat Sheet

## 1. LIST (Ordered, Mutable, Allows Duplicates)

- Create: `my_list = [1, 2, 3]` – Lists store multiple items in order.
- `append(x)` – Adds a single element at the end of the list.
- `extend(iterable)` – Adds multiple elements from another iterable.
- `insert(i, x)` – Inserts element `x` at index position `i`.
- `remove(x)` – Removes the first occurrence of value `x`.
- `pop()` – Removes and returns the last element (or given index).
- `index(x)` – Returns index of first occurrence of `x`.
- `count(x)` – Returns number of times `x` appears.
- `sort()` – Sorts the list in ascending order.
- `reverse()` – Reverses the order of elements.
- `clear()` – Removes all elements from the list.
- `copy()` – Returns a shallow copy of the list.

## 2. TUPLE (Ordered, Immutable, Allows Duplicates)

- Create: `my_tuple = (1, 2, 3)` – Similar to list but immutable.
- `count(x)` – Returns number of times `x` appears in tuple.
- `index(x)` – Returns index of first occurrence of `x`.
- Immutable – You cannot add, remove, or modify elements after creation.

## 3. SET (Unordered, Mutable, No Duplicates)

- Create: `my_set = {1, 2, 3}` – Stores unique elements only.
- `add(x)` – Adds a single element to the set.
- `update(iterable)` – Adds multiple elements from iterable.
- `remove(x)` – Removes element `x` (error if not found).
- `discard(x)` – Removes element safely (no error if missing).
- `pop()` – Removes and returns a random element.
- `clear()` – Removes all elements.
- `union()` – Returns all unique elements from both sets.
- `intersection()` – Returns common elements between sets.
- `difference()` – Returns elements in first set but not in second.
- `symmetric_difference()` – Returns elements not common in both sets.

## 4. DICTIONARY (Key-Value Pairs, Mutable, Unique Keys)

- Create: `my_dict = {'a':1, 'b':2}` – Stores data as key-value pairs.
- `dict[key] = value` – Adds a new key or updates existing key.

- `get(key)` – Returns value safely (None if key not found).
- `keys()` – Returns all dictionary keys.
- `values()` – Returns all dictionary values.
- `items()` – Returns key-value pairs as tuples.
- `update()` – Merges another dictionary into current one.
- `pop(key)` – Removes key and returns its value.
- `popitem()` – Removes and returns the last inserted key-value pair.
- `clear()` – Removes all key-value pairs.
- `copy()` – Returns a shallow copy of the dictionary.